

MANOHAR AKULA

Lead Data Scientist - Supply Chain Analytics, GenAI, and Cloud Operations

Ann Arbor, MI | (602) 815-4823 | akulamanohar96@gmail.com | linkedin.com/in/akulamanohar | Open to relocation

PROFESSIONAL SUMMARY

Lead Data Scientist specializing in supply chain analytics, Generative AI, demand forecasting, and cloud-native data platforms. Builds decision systems for freight cost intelligence, logistics finance, invoice audit quality, executive visibility, and AI-assisted operational analytics. Proven impact includes \$40M+ in avoided logistics costs, \$10M+ projected savings from an AI assistant, 95%+ invoice auto-match, and 2+ TB migrated from legacy formats into BigQuery.

CORE STRENGTHS

- Supply chain analytics, demand and supply forecasting, inventory optimization, logistics cost intelligence, and operations finance.
- GenAI applications, RAG, ReAct agents, LLM evaluation, prompt engineering, and governed SQL generation.
- Cloud-native data engineering with GCP, BigQuery, Cloud Run, Dataform, GCS, Pub/Sub, and pipeline validation.
- Forecasting, anomaly detection, XGBoost, optimization modeling, time-series analysis, and cross-functional executive dashboards.
- Migration automation from proprietary analytics tooling to maintainable, tested, reusable data products.

PROFESSIONAL EXPERIENCE

Ford Motor Company - Lead Data Scientist, Supply Chain

Oct 2023 - Present | GenAI, Python, GCP, BigQuery, Cloud Run, Dataform

- Launched a GenAI freight cost assistant using ReAct agent architecture with RAG, governed SQL generation, dry-run validation, and LLM-as-judge evaluation at 97% accuracy, reducing analyst lookup time by 60% and projecting \$10M+ in operational savings through faster decision-making and standardized cost insights.
- Built ocean accessorial cost analytics for detention, demurrage, dwell-time forecasting, anomaly detection, and root-cause analysis, contributing to approximately \$40M+ in avoided logistics costs.
- Owned logistics finance accrual automation with estimate-vs-actual variance tracking, deduplication, monthly journal entry support, and SOX-aware validation workflows.
- Led freight audit analytics and discrepancy recovery, raising invoice auto-match above 95% and reducing duplicate or incomplete invoice records by 90%+.
- Created director-level premium freight dashboards with aging metrics, predictive analytics, data gap resolution, and executive-ready visibility into high-priority freight spend.
- Built an award-winning agentic AI migration framework for sunsetting legacy Alteryx workflows, automating conversion to BigQuery Standard SQL with AST validation, supported by 494+ automated tests, and recognized with a Ford innovation award for accelerating enterprise cloud migration.
- Built an AI Dashboard Migrator converting QlikSense dashboards to production-ready Python Dash code, eliminating millions in annual licensing costs, and engineered a QVX-to-BigQuery migration pipeline using streaming parallel processing, proprietary binary decoding, and configurable worker pools to migrate 2+ TB of legacy data.
- Standardized multiple freight analytics pipelines on GCP-native tooling with Dataform/SQLX, automated validation, lineage, and reusable downstream data foundations.

Blue Guardian - Data Scientist / Data Engineer

Jan 2023 - Oct 2023 | NLP, Python, Keras, TensorFlow, AWS, OpenAI

- Built AWS-based ML workflows using S3 and SageMaker to train, host, and deploy models for mental health data analysis.
- Developed a LangChain and OpenAI-powered dataframe agent that improved depression detection precision to 83%.
- Integrated CNN and BERT NLP frameworks, improving model accuracy by 20% through stronger text representation and model architecture.

Arizona State University - Data Science Consultant, Technology Office

Dec 2021 - Dec 2022 | GCP, ETL, Python, SQL, AWS

- Increased student enrollment by 20% and revenue by 10% through predictive modeling and analytics dashboards.
- Implemented ETL and ML pipelines with Cloud Dataproc, PySpark, BigQuery, Vertex AI, Redshift, Lambda, Pub/Sub, and Cloud Functions.

Accucorps Technologies Pvt Ltd - Data Scientist

Jul 2017 - Jul 2021 | Azure, Spark, Power BI

- Improved purchase prediction accuracy by 30% using deep-learning classification and regression models.
- Automated Azure Data Factory pipelines and Power BI dashboards for real-time business intelligence.

SELECTED SUPPLY CHAIN AND AI SYSTEMS

- Freight Cost AI Assistant: natural-language analytics over large freight cost data with SQL generation, retries, telemetry, and evaluation controls.
- Freight AI Knowledge Architecture: semantic-layer vision connecting freight dashboards, chatbots, process flows, and analytics products into an agent-queryable knowledge graph.
- QlikSense-to-Dash Converter: automated conversion of proprietary dashboards into Python Dash applications, eliminating millions in annual licensing costs.
- Dataform Migration Engine: repeatable conversion and validation pattern for moving Alteryx ETL workflows into cloud-native SQLX pipelines.

TECHNICAL SKILLS

- Languages: Python, SQL, BigQuery Standard SQL, Dataform SQLX, R, C++.
- AI and Agents: OpenAI GPT, Gemini, LangChain, RAG, ReAct agents, planner-writer workflows, prompt engineering, LLM evaluation.
- Machine Learning: XGBoost, Scikit-learn, TensorFlow, PyTorch, Keras, BERT, CNNs, regression modeling, time-series forecasting.
- Cloud and Data: GCP, BigQuery, Vertex AI, Cloud Run, Cloud Run Jobs, Dataform, GCS, Pub/Sub, Secret Manager, AWS, Azure.
- Data and BI: Pandas, NumPy, PySpark, Alteryx, Qlik Sense, Dash/Plotly, Power BI, Tableau.

EDUCATION

- M.S. in Artificial Intelligence and Robotics - Arizona State University, Tempe, AZ.
- B.Tech in Electrical and Electronics Engineering - Jawaharlal Nehru Technological University, Hyderabad, India.